Notes: Koijen & Yogo (AER: Insights, 2023)

Understanding the Ownership Structure of Corporate Bonds

Contents

1 Big picture

- **Question.** Why are insurers the largest institutional holders of corporate bonds, even though a standard insurance model with risk-based capital regulation predicts they should prefer riskless bonds?
- **Core puzzle.** Insurers hold a lot of credit risk, and in particular tilt toward highly rated investment-grade corporate bonds instead of Treasuries. Standard theory says this should not happen if risky assets require capital yet offer no abnormal risk-adjusted return.
- **Paper's answer.** Put insurers into an equilibrium asset-pricing model with leverage-constrained households and institutional investors. Once leverage constraints flatten the security market line, low-beta assets earn positive alpha. Insurers have relatively cheap access to leverage through underwriting, so they optimally hold a leveraged portfolio of low-beta assets such as investment-grade corporate bonds.
- **Empirical facts the model is built to explain.** Insurers dominate U.S. corporate bond ownership; both life and property–casualty insurers hold more corporate bonds than Treasuries; within corporate bonds they tilt toward safer ratings; and after the global financial crisis the credit risk of property–casualty portfolios rises relative to life insurers.
- **Main theoretical mechanism.** Other investors are leverage constrained and therefore overvalue levered exposure to low-beta assets. Insurer equity provides exactly that exposure. Insurers exploit this by holding low-beta assets and issuing liabilities through their underwriting business.
- **Main implication.** Insurers are not simply "reaching for yield." They are performing an intermediation role: they absorb low-beta assets and relax the leverage constraints of other investors.
- **Why this matters.** The paper links insurance-sector portfolio choice to corporate funding conditions, intermediary asset pricing, and the low-beta anomaly. It also gives a unified way to think about why regulation, downgrades, and liability structure matter for bond demand.

2 Data and measurement (key objects)

2.1 What the non-theory part of the paper does

- **Nature of evidence.** The paper is primarily a short theory paper, disciplined by four descriptive figures rather than regression tables.
- **Role of the data.** The descriptive evidence motivates the puzzle and anchors the model's comparative statics.

2.2 Institutional ownership of U.S. corporate bonds

- **Source.** Financial Accounts of the United States, 1945–2017.
- **Key variable.** Share of U.S. corporate bonds held by insurers, pension funds, banks, mutual funds, and government-sponsored enterprises.
- **Main fact.** Insurers are the largest institutional holders of U.S. corporate bonds throughout the postwar period; in 2017 they hold 38%.

2.3 Insurers' bond portfolio composition

- **Source.** National Association of Insurance Commissioners, annual long-term bond holdings, 1994–2019.
- **Breakdown.** U.S. Treasury bonds, U.S. agency bonds, publicly traded corporate bonds, other U.S. government bonds, local and foreign government bonds, and private corporate bonds.
- **Main fact.** Life insurers allocate most of their bond portfolio to public and private corporate bonds; property–casualty insurers also move from a Treasury-heavy mix toward corporate bonds over time.
- **Interpretation.** Because liabilities are not directly exposed to credit risk, these bond holdings create credit-risk mismatch.

2.4 Credit-quality composition within corporate bonds

- **Source.** Same
  NAIC holdings data, merged to Mergent bond information.
- **Rating buckets.**
  NAIC 1 is safest and
  NAIC 6 riskiest.
- **Main fact.** Insurers overweight
  NAIC 1 and especially
  NAIC 2 bonds relative to the market portfolio, and underweight
  NAIC 3–6 bonds.
- **Interpretation.** This is exactly the paper's notion of a tilt toward low-beta assets: not "junk," but relatively safe corporate bonds.

2.5 Portfolio credit risk

- **Measure.** Weighted-average ten-year cumulative default rate, obtained by mapping ratings to default probabilities.
- **Main fact.** Life insurers keep portfolio credit risk broadly stable at roughly 2–4%, while property–casualty insurers increase it from about 0.9% in 1994 to 2.8% in 2019.
- **Interpretation.** The cross-insurer divergence after the global financial crisis is one of the main comparative-static targets of the model.

3 Models and empirical specifications (all equations + interpretation)

3.1 Annuities and the insurance-sector source of leverage

Households survive to period 1 with probability π and can buy annuities from insurers. The annuity gross return conditional on survival is

$$R_L = \frac{1}{P_L} = \left(1 - \frac{1}{\varepsilon}\right)\frac{R_f}{\pi}. \tag{1}$$

- **Interpretation.** P_L is the annuity price in period 0 and ε is the demand elasticity for annuities.
- **Economic meaning.** Insurers earn underwriting profits because annuities are sold at a markup over actuarially fair value.
- **Why it matters.** This underwriting activity is what gives insurers relatively cheap access to leverage.

3.2 Risky-asset dividend structure

Other firms' dividends follow a one-factor structure:

$$d = \mathbb{E}[d] + \beta F + \nu, \tag{2}$$

with covariance matrix

$$\mathrm{Var}(d) = \sigma_F^2 \beta\beta' + \mathrm{diag}(\sigma_\nu^2). \tag{3}$$

- **Interpretation.** β is the vector of market betas, F is the common factor, and ν is idiosyncratic risk.
- **Why beta matters.** The whole mechanism of the paper rests on the fact that low-beta assets earn positive alpha when other investors cannot lever up enough.

3.3 Insurer balance sheet and regulatory cost

Let x_I denote insurers' holdings of risky assets. Required capital is proportional to the risk-weighted norm of that portfolio:

$$K_I^{req}(x_I) = \sqrt{x_I' \exp(\phi\beta) x_I}, \tag{4}$$

and the cost of regulatory frictions is quadratic in required capital:

$$C(x_I) = \frac{1}{2} x_I' \exp(\phi\beta) x_I. \tag{5}$$

- **Interpretation.** $\exp(\phi\beta)$ is the diagonal matrix of risk weights. Higher-beta assets require more capital.
- **Role of ϕ.** A higher ϕ means effective capital regulation is more sensitive to risk. The paper also interprets higher ϕ as tighter insurer-specific financial frictions, especially for life insurers with risky liabilities.

Insurers' period-0 assets are

$$A_{I,0} = E - C(x_I) + P_L Q, \tag{6}$$

and their dividends in period 1 are

$$d_I = d'x_I + R_f(A_{I,0} - p'x_I) - \pi Q = d'x_I + R_f[E - C(x_I) - p'x_I] + (R_f P_L - \pi)Q. \tag{7}$$

Using the optimal annuity markup, this becomes

$$d_I = d'x_I + R_f[E - C(x_I) - p'x_I] + \frac{\pi Q}{\varepsilon - 1}. \tag{8}$$

- **Interpretation.** Insurer value has two moving parts before equilibrium asset pricing enters: portfolio payoffs and underwriting profits.
- **Key point.** Even before talking about equilibrium asset prices, riskier assets tighten the regulatory constraint and therefore lower firm value through $C(x_I)$.

3.4 Household problem

Households allocate wealth between risky assets and annuities:

$$A_{H,0} = P'X_H + P_L Q, \tag{9}$$

with period-1 wealth conditional on survival

$$A_{H,1} = D'X_H + Q = D'X_H + R_L(A_{H,0} - P'X_H). \tag{10}$$

They maximize mean-variance utility

$$\mathbb{E}[A_{H,1}] - \frac{\gamma_H}{2} \operatorname{Var}(A_{H,1}) \tag{11}$$

subject to a leverage constraint

$$P'X_H \leq A_{H,0}. \tag{12}$$

The first-order condition gives

$$X_H = \frac{1}{\gamma_H} \Sigma^{-1}[\mu - (R_L + \lambda_H)P]. \tag{13}$$

- **Interpretation.** A binding household leverage constraint appears as an extra shadow cost λ_H.
- **Economic meaning.** The leverage constraint raises the effective hurdle rate households apply to risky assets.

3.5 Other institutional investors

Institution j has period-1 wealth

$$A_{j,1} = D'X_j + R_f(A_{j,0} - P'X_j), \tag{14}$$

maximizes

$$\mathbb{E}[A_{j,1}] - \frac{\gamma_j}{2} \operatorname{Var}(A_{j,1}), \tag{15}$$

subject to leverage

$$P' X_j \leq \frac{A_{j,0}}{\omega_j}. \tag{16}$$

Optimal risky demand is

$$X_j = \frac{1}{\gamma_j} \Sigma^{-1} \big[\mu - (R_f + \lambda_j) P\big]. \tag{17}$$

- **Interpretation.** ω_j controls how tightly institution j is levered, and λ_j is the associated shadow value of relaxing that constraint.
- **Economic meaning.** These investors are the reason the security market line is too flat and low-beta assets earn positive alpha.

3.6 Market clearing and asset prices

Risky-asset market clearing is

$$X_I + X_H + \sum_{j=1}^{J} X_j = \mathbf{1}. \tag{18}$$

Combining optimal demands yields equilibrium asset prices

$$P = \frac{1}{R + \lambda} \big[\mu - \gamma \Sigma (\mathbf{1} - X_I)\big], \tag{19}$$

where the paper defines aggregate risk-tolerance and shadow-cost objects by

$$\frac{1}{\gamma} = \frac{1}{\gamma_H} + \sum_{j=1}^{J} \frac{1}{\gamma_j}, \qquad R = \frac{\gamma}{\gamma_H} R_L + \sum_{j=1}^{J} \frac{\gamma}{\gamma_j} R_f, \qquad \lambda = \frac{\gamma}{\gamma_H} \lambda_H + \sum_{j=1}^{J} \frac{\gamma}{\gamma_j} \lambda_j. \tag{20}$$

The price of outside risky firms simplifies to

$$p = \frac{1}{R + \lambda} \big(\mathbb{E}[d] - \gamma \operatorname{Var}(d) \mathbf{1}\big). \tag{21}$$

- **Interpretation.** The entire effect of insurer portfolio choice on other firms' prices runs through the aggregate leverage-pressure term λ.
- **Key insight.** Insurers matter for asset prices by relaxing other investors' leverage constraints, not because they directly price-discriminate across all securities.

The insurers' own equity value can be decomposed into three sources:

$$p_I = \frac{1}{R+\lambda}\left[\underbrace{\frac{\lambda+R-R_f}{R_f}\big(\mathbb{E}[d]-\gamma\,\mathrm{Var}(d)\mathbf{1}\big)'x_I}_{\text{portfolio choice}} + \underbrace{R_f\big(E-C(x_I)\big)}_{\text{regulatory frictions}} + \underbrace{\frac{\pi Q}{\varepsilon-1}}_{\text{underwriting profits}}\right]. \tag{22}$$

- **Interpretation.** This is the paper's main value decomposition.
- **Economic meaning.** Insurer value comes from: (i) holding the right risky assets; (ii) economizing on regulatory capital costs; and (iii) extracting markups in annuity underwriting.
- **Special benchmark.** If leverage constraints do not bind ($\lambda = 0$) and the effective discount rate equals the riskless rate ($R = R_f$), the portfolio-choice term disappears and the standard insurance result is restored.

3.7 Optimal insurer portfolio and the low-beta tilt

Maximizing insurer value with respect to x_I yields

$$x_I = \frac{\lambda+R-R_f}{R_f(R+\lambda)}\exp(-\phi\beta)\big[\mathbb{E}[d]-\gamma\,\mathrm{Var}(d)\mathbf{1}\big]. \tag{23}$$

Using the factor structure, this can be written as

$$x_I = \frac{\lambda+R-R_f}{R_f(R+\lambda)}\exp(-\phi\beta)\big[\mathbb{E}[d]-\gamma\sigma_F^2\beta\beta'\mathbf{1}-\gamma\sigma_\nu^2\big]. \tag{24}$$

The core comparative static is

$$\frac{\partial x_I(n)}{\partial\beta(n)} < 0. \tag{25}$$

- **Interpretation.** Insurers strictly tilt away from higher-beta assets and toward lower-beta assets.
- **Economic meaning.** In an economy with a flattened security market line, low-beta assets have positive alpha; insurers lever them up.
- **Standard-theory case.** When $\lambda = 0$ and $R = R_f$, the coefficient in front is zero, so $x_I = 0$ and insurers hold only the riskless asset.

A second comparative static links regulation to portfolio choice:

$$\frac{\partial^2 x_I(n)}{\partial\beta(n)\,\partial\phi} < 0 \qquad \text{for sufficiently low-beta assets.} \tag{26}$$

- **Interpretation.** When effective capital regulation becomes more sensitive to risk, insurers retreat especially strongly from assets whose beta rises, such as downgraded bonds.
- **Empirical mapping.** This is the comparative static the paper uses to rationalize insurer sales of downgraded bonds and the differential behavior of life versus property–casualty insurers.

4 Identification and instruments (what makes the estimates credible)

4.1 No econometric identification design

- **Important clarification.** This paper does not estimate causal coefficients with an IV, diff-in-diff, or structural micro estimation. It is a theory paper plus descriptive facts.
- **So what is being identified?** The paper identifies a *mechanism*: leverage-constrained investors create a low-beta anomaly, and insurers with cheap leverage optimally absorb low-beta assets.

4.2 How the paper disciplines the mechanism

- **Fact 1.** Insurers hold a lot of corporate bonds even though standard insurance theory predicts riskless bonds.
- **Fact 2.** Within corporate bonds, insurers tilt toward safer ratings, not toward the riskiest assets.
- **Fact 3.** Property–casualty insurers increase credit risk relative to life insurers after the global financial crisis.
- **Discipline.** The model must explain all three facts jointly. A simple "reaching-for-yield" story would struggle with Fact 2, because insurers concentrate in low-beta investment-grade bonds rather than high-beta junk.

4.3 Comparative-statics mapping to outside evidence

- **Downgrades.** A bond downgrade can be read as an increase in beta; then the model predicts insurers sell more of it, especially when effective capital pressure ϕ is high.
- **MBS after the crisis.** If capital rules stop penalizing risk for a certain asset class, that is like lowering ϕ for that class; insurers no longer have the same incentive to dump downgraded assets.
- **Operating losses.** A hit to insurer balance sheets can tighten effective risk constraints, again behaving like a higher ϕ.
- **Low-rate environment.** When other investors become more leverage constrained, λ rises and insurers have even more incentive to hold risky low-beta assets. But when life insurers' own liability-side fragility tightens ϕ, their risk-taking can decline relative to property–casualty firms.

4.4 What makes the story persuasive

- The mechanism is **internally coherent**: it reproduces the standard no-corporate-bond prediction as a limiting case.
- It is **asset-pricing consistent**: the low-beta anomaly is the wedge that makes insurer portfolio choice privately valuable.
- It is **cross-sectionally sharp**: safer corporate bonds should be exactly where insurers concentrate.
- It is **institutionally grounded**: underwriting liabilities provide a concrete source of cheap leverage that other investors lack.

5 Tables and figures

5.1 Figure 1: Institutional ownership of corporate bonds

Read:

- Insurers are the dominant institutional holders of U.S. corporate bonds over the full postwar period.
- In 2017, insurers hold 38%, versus 30% for mutual funds, 16% for pension funds, and 10% for banks.
- This figure motivates the whole paper: whatever the correct theory is, it must explain why insurers, not some other intermediary, sit at the center of this market.

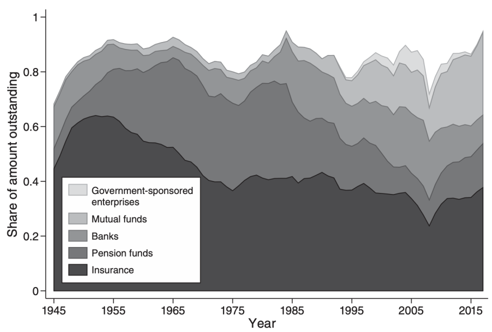

Figure 1. Institutional Ownership of Corporate Bonds

Note: Authors' tabulation based on the *Financial Accounts of the United States* for 1945–2017 (Board of Governors of the Federal Reserve System 2017).

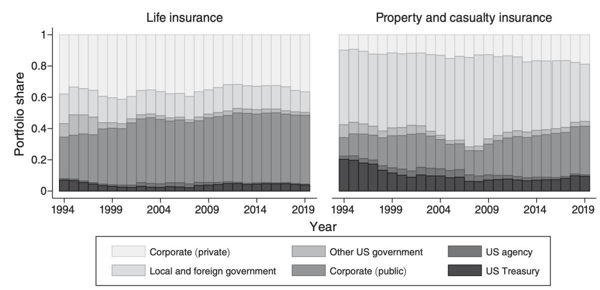

Figure 2. Portfolio Composition

Notes: The long-term bond holdings are from the National Association of Insurance Commissioners (1994–2019; Schedule D, Part 1). The bottom three categories (US Treasury, US agency, and publicly traded corporate bonds) represent publicly traded and non-asset-backed securities with coverage in Mergent (2021). The top three categories include asset-backed securities, mortgage-backed securities, and private placement bonds.

5.2 Figure 2: Portfolio composition

Read:

- Life insurers hold most of their bond portfolios in public and private corporate bonds rather than Treasuries.
- Property–casualty insurers start out more Treasury-heavy in the 1990s but gradually rotate toward corporate bonds.
- The key point is not just that insurers buy risky assets, but that they do so in size despite capital regulation and liability structures that seem, at first glance, to discourage it.

5.3 Figure 3: Corporate bond portfolio composition by NAIC bucket

Read:

- Insurers overweight NAIC 1 and especially NAIC 2 bonds, and underweight NAIC 3–6 bonds relative to the market portfolio.
- After 2007 there is a broad migration in the market from NAIC 1 toward NAIC 2, consistent with the growth of the BBB segment.

- This is central for the theory: insurers are not chasing the highest yields mechanically; they tilt toward *low-beta corporate credit.*

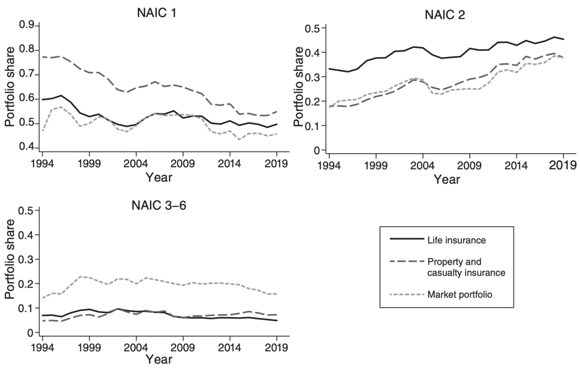

Figure 3. Corporate Bond Portfolio Composition

Notes: This figure shows the corporate bond portfolio share by NAIC designation for life insurers and property and casualty insurers. It also shows the market portfolio weights for the universe of corporate bonds that are held by the insurance sector. The long-term bond holdings are from the National Association of Insurance Commissioners (1994–2019; Schedule D, Part 1). The sample consists of corporate bonds in Mergent (2021) that are publicly traded and not asset backed.

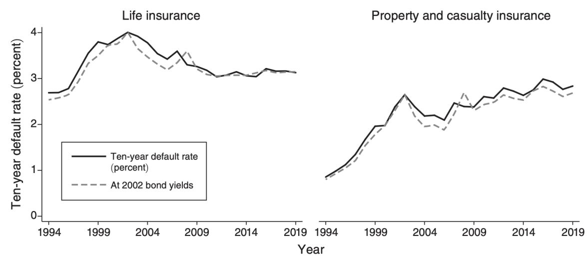

Figure 4. Credit Risk of Bond Portfolios

Notes: This figure shows the weighted average ten-year cumulative default rate of bond portfolios for life insurers and property and casualty insurers. The long-term bond holdings are from the National Association of Insurance Commissioners (1994–2019; Schedule D, Part 1). The sample consists of US Treasury, US agency, and corporate bonds in Mergent (2021) that are publicly traded and not asset backed. The ten-year cumulative default rate is assigned to each bond based on the median of its S&P, Moody's, and Fitch ratings. An alternative calculation of the weighted average ten-year cumulative default rate fixes bond yields by maturity and credit rating to their values in 2002.

5.4 Figure 4: Credit risk of bond portfolios

Read:

- Life insurers maintain relatively stable portfolio credit risk over time, around 2–4% in ten-year default-rate terms.
- Property–casualty insurers steadily raise credit risk, from about 0.9% to 2.8%.
- The alternative line holding yields fixed at 2002 values shows the rise is not merely a price effect; portfolio composition and rating buckets change materially.
- This figure motivates the paper's final comparative-static discussion of why property–casualty insurers move more into credit risk than life insurers after the global financial crisis.

6 Short summary

- **Conclusion.** Insurers hold corporate bonds because they have a comparative advantage in levering low-beta assets when other investors face leverage constraints.
- **Intuition.** Households and institutions want more leveraged exposure to low-beta assets than they can create themselves. Insurer equity gives them that exposure, and insurers implement it by holding investment-grade corporate credit.
- **Empirical lesson.** The insurance sector's bond demand is best understood through equilibrium asset-pricing and liability-side leverage, not through a partial-equilibrium "why do insurers like credit risk?" lens alone.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** A standard insurance model predicts riskless bond holdings; an equilibrium model with leverage-constrained outside investors predicts the observed tilt toward investment-grade corporate bonds.
- **Magnitude.** The paper is not about a single estimated coefficient; its quantitative punch comes from matching large portfolio shares and persistent industry-wide holdings patterns.
- **Credibility.** The model explains simultaneously why insurers hold corporate bonds, why they prefer safer corporate bonds, and why property–casualty insurers become relatively riskier than life insurers after the crisis.
- **Mechanism evidence.** The comparative statics line up with prior evidence on downgrades, regulatory pressure, and insurer portfolio rebalancing.
- **Broader implication.** Insurers are a special intermediary because underwriting liabilities give them a form of cheap leverage that can be transformed into demand for low-beta assets.
- **Policy implication.** Regulation and liability design matter not only for insurer solvency but also for asset prices and corporate financing conditions in the broader economy.

7.2 Economic intuition

- The low-beta anomaly means low-beta securities offer unusually high risk-adjusted returns relative to what constrained investors can achieve on their own.
- Insurers can issue liabilities through underwriting, lever those low-beta assets, and sell claims on that levered bundle via their equity.
- Risk-based capital regulation pushes the portfolio toward the safest assets within corporate credit, which is why insurers favor investment-grade bonds rather than the riskiest debt.
- Differences in liability fragility and capital pressure explain why life insurers and property–casualty insurers do not make the same portfolio choices over time.

7.3 Takeaway

This is a short paper with a clean idea: corporate bond ownership is an equilibrium outcome of insurer balance-sheet technology and outside investors' leverage constraints. Once low-beta assets carry positive alpha, the insurance sector naturally becomes the main holder of investment-grade corporate credit. The most useful way to remember the paper is that insurers are not just buyers of bonds; they are suppliers of levered low-beta exposure to the rest of the financial system.

7.4 Potential extensions highlighted by the authors

- **Interest-rate risk mismatch.** The paper abstracts from maturity choice, but in reality insurers trade off credit premia against duration management.

- **Capital structure.** If insurers can issue public debt or alter payout, leverage choice itself becomes endogenous.
- **Insurance pricing.** If policies are subject to capital requirements or default risk, underwriting and portfolio choice interact more tightly.
- **Agency problems.** Risk shifting, limited liability, and guaranty funds may enrich the portfolio-choice problem further.